MILITARY SPECIFICATION

BARRIER AIRCRAFT ARRESTING

This specification is approved for use within the Department of the Air Force and is available for use by all departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable, rotary, hydraulic, barrier aircraft arresting system, designated BAK-13A/F48A.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

*	A-A-1830	Tape Pressure Sensitive Adhesive Box Closure
*	TT-V-121	Varnish, Spar, Water Resisting
*	UU-T-81	Tag, Shipping and Stock
*	PPP-B-601	Box, Wood, Cleated Plywood
*	PPP-B-621	Box, Wood, Nailed and Lock-Corner
*	PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
*	PPP-P-40	Packaging and Packing of Hand Tools

MILITARY

*	MIL-C-104	Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted
*	MIL-P-116	Preservation, Methods of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1710

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	MIL-P-514	Plate, Identification, Instruction and Marking, Blank
	MIL-G-8402	Gages, Pressure, Dial Indicating, General Specification for
*	MIL-E-10062	Engines, Preparation for Shipment and Storage of
*	MIL-E-17555	Electronic and Electrical Equipment, Accessories and Provisioned Items (Repair Parts) Packaging of
	MIL-W-38461	Webbing, Nylon, Aircraft Arresting

STANDARDS

FEDERAL

	FED-STD-H28	Screw Thread Standards for Federal Services

MILITARY

	MIL-STD-12	Abbreviations for Use on Drawings, Specifications, Standards, and in Technical Documents
	MIL-STD-17	Mechanical Symbols
	DOD-STD-100	Engineering Drawing Practices
*	MIL-STD-129	Marking for Shipment and Storage
	MIL-STD-130	Identification Marking of U.S. Military Property
	MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
*	MIL-STD-147	Palletized Unit Loads
*	MIL-STD-163	Steel Mill Products, Preparation of Shipment and Storage
	MIL-STD-808	Finish, Materials and Processes for Corrosion Prevention and Control in Support Equipment
	MIL-STD-810	Environmental Test Methods and Engineering Guidelines
	MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
	MIL-STD-831	Test Reports, Preparation of
	MIL-STD-889	Dissimilar Metals
*	MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and with Appropriate Test Methods
*	MIL-STD-2073	DOD Materiel Procedures for Development and Application of Packing Requirements

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Air Force

71211	Hardware Kit Assembly, Anchoring and Leveling Hardware, BAK-13A Arresting System Foundations
71214	BAK-13A Arresting System Foundations, Structural Plans

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Naval Air Engineering Center (Code 80020)

510445	Tape Connector Assembly
515053	Pendant Assembly

E. W. Bliss Company

52-E-182 & Drawing List P-2154	General Arrangement BAK-13A Arresting Engine

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting acvitity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

Naval Air Engineering Center

NAEC Mis. No 08787	Wire Rope, Steel, 1-1/4 Inch Diameter (12 x 6/6 x 30) Nonrotating, Polypropylene Core, High Strength (for Aircraft Arresting) (Rev B or later issue)

(Application for copies should be addressed to the Naval Air Engineering Center, (SSIED), Code 91, Lakehurst, N.J. 08733-5092.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.4).

3.2 Components. The barrier aircraft arresting system shall consist of the following major components:

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Item or Component Name	See Requirement
Energy absorber	3.9
Purchase element tapes	3.10
Purchase element tape connectors	3.11
Hook cable	3.12
Runway edge sheaves	3.13
Fairlead tubes and couplings	3.14
Special service tools and kits	3.15

3.3 General Requirements. The requirements of Drawing 52-E-182 and details thereto as shown on Drawing list P2154 (see 6.2) apply as requirements of this specification with the exceptions and additions specified herein.

3.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.5 Materials

3.5.1 Metals. Corrosion-resistant metals shall be used in the construction of the aircraft arresting system wherever protective finishing is not practical. Nonexposed lubricated metal parts and surfaces need not be given a protective coating and need not be corrosion-resistant. Protective coatings shall be as specified herein.

3.5.2 Dissimilar metals. Insofar as practicable, the use of dissimilar metals in contact with each other shall be avoided. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces will be permitted. The use of dissimilar metals separated by insulating material will be permitted. Dissimilar metals are defined in MIL-STD-889.

3.6 Design. The aircraft arresting system shall be a portable, velocity-sensitive, turbine-type, rotary hydraulic system based on the split absorber concept. The energy absorber units shall be sited on opposite sides of the runway with no requirement for cross-runway interconnection other than the hook cable. The aircraft arresting system shall be designed and equipped to provide a normal system runout of 950 feet. The aircraft arresting system shall be designed for soil surface, surface type concrete foundation, or permanent pit type installation. Installation shall be accomplished in the split configuration with runway edge sheave assemblies sited adjacent to the runway edges and the energy absorber units installed outboard from the sheave assemblies at split distances varying from 60 to 150 feet. The procuring activity shall specify the following for each site where the aircraft arresting system is to be used (see 6.2):

a. Type of installation (see 3.6.1, 3.6.2, and 3.6.3)

b. Length of hook cable (see 3.12)

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c. Span between runway edge sheaves (see 3.10, 3.12, and 6.3.2)

d. Split distance between the runway edge sheaves and energy absorber (see 3.10, 3.12, and 6.3.3)

e. Items, components, or foundations that will be furnished by the government (see 6.2)

The aircraft arresting system shall be capable of accommodating at least 1,000 arrestments at the 30- to 40-million foot-pound energy level without replacement of any major assembly or component, except for consumables such as purchase element tapes, purchase element tape connectors, and hook cables. System down-time required for replacement of consumable components shall be kept to a minimum.

3.6.1 Soil surface installation. The design of the aircraft arresting system shall permit installation on a graded soil surface using aluminum stakes and deadman anchors. The effort required to install the complete system shall not exceed 100 man-hours, and elapsed time shall not exceed 10 clock hours. Arresting systems for soil surface installations shall be in accordance with Drawing 52-E-182 and shall consist of at least the following major components, arranged essentially as shown thereon. Additional components, including hardware, required to complete the aircraft arresting system for this particular type installation shall be as shown on Drawing 52-E-182 and Drawing list P-2154.

Component	See Paragraph	Quanity Required
Energy absorber	3.9	2
Purchase element tapes	3.10	2
Purchase element tape connectors	3.11	2
Hook cable	3.12	1
Runway edge sheaves	3.13	2
Fairlead tubes	3.14	as specified
Special service tools and kits	3.15	1 set.

3.6.1.1 Reinstallation kit. When specified (see 6.2), a reinstallation kit shall be provided for each soil surface installation. The kit shall contain the anchoring hardware for the absorber. The kit shall permit quick removal and reinstallation of the arresting system by avoiding the need to dig up anchoring hardware. Necessary hardware is shown on drawing 52-B-2518.

3.6.2 Installation, surface type concrete foundations. The design of the aircraft arresting system shall be adaptable to installation on a concrete foundation conforming to Drawing 71214 using a hardware kit assembly in accordance with Drawing 71211 in place of the stakes and deadman anchors shown on Drawing 52-E-182. (Drawing 71214 replaces Drawing 70WA1049 referenced

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on Drawing 52-E-182.) When specified (see 6.2), a concrete foundation in accordance with Drawing 71214 shall be provided. Installation shall be in accordance with Drawing 52-E-182 and shall consist of at least the following major components, arranged essentially as shown thereon. Additional components, including hardware, required to complete the aircraft arresting system for this particular type installation shall be as shown on Drawing 52-E-182 and Drawing list P-2154, except as modified by use of Drawings 71211 and 71214.

Component	See Paragraph	Quantity Required
Energy absorber	3.9	2
Purchase element tapes	3.10	2
Purchase element tape connectors	3.11	2
Hook cable	3.12	1
Runway edge sheaves	3.13	2
Fairlead tubes	3.14	as specified
Special service tools and kits	3.15	1 set

3.6.3 Installation, permanent, pit type. The design of the aircraft arresting system shall be adaptable to installation in an underground concrete pit having a top which is substantially flush with the surrounding terrain. In the event that pit type installations are required, necessary drawings and specifications will be furnished by the procuring activity (see 6.2).

3.6.4 Reliability. The aircraft arresting system shall have a reliability (probability of success) of not less than 0.97 at a 0.90 confidence level. In this regard, the system shall be capable of not less then 76 consecutive arrestments without failure considering any combination of performance parameters specified herein and in accordance with proper maintenance procedures. Failures of major components or parts of a major component (excluding purchase tape, hook cable, and purchase element tape connectors) which render the component or system ineffective for the next arrestment shall constitute a basis for rejection of the system (see 6.3.1).

3.6.5 Maintainability. The barrier aircraft arresting system shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure

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c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to approval by the procuring activity

e. Minimum number of tools required for maintenance by design practice such as reducing the variety of bolthead sizes to the practicable minimum

f. Consumable spare parts (except purchase element tapes, hook cable, and purchase element connectors) which require more than 10 minutes to replace shall have a minimum service life of 76 arrestments.

3.6.5.1 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.7 Construction. The aircraft arresting system shall be so constructed that no parts will work loose in service. It shall be built to withstand the loads, strains, vibrations, and other conditions incident to service, shipping, and storage. (Reference to parts working loose in service includes premature functioning of parts, devices, or mechanisms which degrade system performance or cause equipment to operate improperly.)

3.8 Performance

3.8.1 Environmental conditions. The aircraft arresting system shall be capable of operating satisfactorily and without damage under the following environmental conditions:

a. Temperatures ranging from -40° to +125°F

b. Humidity up to 100 percent

c. Fungus

d. Rain

e. Salt-fog

f. Dust.

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3.8.2 Operational arrestments. The aircraft arresting system shall meet or exceed the following performance requirements with regard to energy absorption in relation to speed/weight for aircraft arrestments:

a. When the arresting system is installed to provide a system runout distance of 950 feet and having a 190-foot runway span, a split distance of 60 feet between runway edge sheaves and the energy absorber and using a 1-1/4 inch-diameter hook cable, the system shall:

(1) Have an energy capacity up to 85 x 10^6 foot-pounds

(2) Withstand hook engagements off-center up to 50 feet at the following weights and speeds:

Aircraft Weight (Pounds)	Engaging Speed (Knots)
20,000	180
30,000	180
40,000	180
50,000	180
60,000	180
70,000	166
80,000	156

(3) Not impose average hook loads upon the aircraft exceeding the following:

Aircraft Weight (Pounds)	Engaging Speed (Knots)	Average Hook Load (Pounds)
25,000	120	38,000
	140	53,000
	160	72,000
	180	93,000
40,000	120	43,000
	140	61,000
	160	82,000
	180	107,000
55,000	120	47,000
	140	66,000
	160	88,000
	180	117,000
70,000	100	37,000
	120	56,000
	140	77,000
	150	88,000

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Aircraft Weight (Pounds)	Engaging Speed (Knots)	Average Hook Load (Pounds)
85,000	100	40,000
	120	60,000
	130	70,000.

(4) Be capable of accepting twenty arrestments per hour

b. Under all conditions above, the tensile load in the purchase system shall not exceed 60 percent of the strength of the purchase system at its weakest points

c. The unbalanced steering force exerted upon an aircraft in off-center hook engagements shall be zero or shall have a moderate tendency to steer the aircraft toward center during runout.

3.9 Energy absorber. The energy absorber shall be of the rotary hydraulic type and shall limit aircraft runout to 950 feet maximum when the span and split distance effect is considered. The absorber shall have sufficient thermal capacity to accommodate not less than 20 arrestments within a 1-hour period. Average energy level for an arrestment will be approximately 40 million foot-pounds. The absorber cooling shall be by integral means and external fluid reservoir. The absorber shall meet the performance requirements specified herein using a water glycol mixture as the hydraulic medium. All hydraulic components shall have suitable factors of safety when working at maximum energy conditions. Pressure gages shall conform to MIL-G-8402. Maximum temperatures resulting from the rapid recycle requirement shall not in any way degrade equipment performance, safety, or reliability. A fluid temperature gage with a tolerance of $\pm 5^{\circ}$F and indices at 5° increments shall be provided and installed on the operator's console located on the retract engine. The gage shall be capable of recording the absorber tub temperature between 0° and 240°F.

3.9.1 Purchase tape storage reels. The tape reel, part No. 52-D-785, shall be designed to achieve maximum strength-to-weight ratio in order to minimize inertia loads. The reel shall be designed and fabricated so as to obtain the maximum size side plate diameter commensurate with the available space in the energy absorber. A side plate diameter of 74 inches will be acceptable.

3.9.2 Arrestor rewind system. The Wisconsin gasoline engine, model V 465 D (65 hp) shall be used to accomplish arresting system rewind. In addition, the power train shall consist of suitable transmission, reduction gear box, backstop clutch, and other elements required to insure proper functioning. The time required to return the system to battery position shall not exceed 1 minute. After return to battery position, the rewind system shall provide 2,900 pounds minimum pretensioning of the hook cable. Provision shall be made for manual rewind in the event of rewind system breakdown.

3.9.3 Retract engaging mechanism. An engaging mechanism shall be supplied to engage the rewind system to the purchase element tape reels in order to rewind the tapes after an arrestment. A back-stopping clutch shall prevent reverse rotation of the tape reels and hold system pretension. A spring-loaded pawl of the engaging mechanism will release the tape reel during an arrestment at a minimum load of 6,000 pounds. Provision shall be made to manually release system pretension for normal maintenance operations.

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3.9.4 Tight wrap roller. A tight wrap roller arm assembly shall be provided to insure a tight purchase element tape wrap on the purchase element tape reels. A controlled force of 1,000 pounds minimum shall be applied to the tape on the reel during the entire rewind operation. Provisions shall be made to disengage the roller arm from the tape reel prior to an arrestment. This mechanism shall contain a hydraulically actuated roller arm that can be controlled from the operators control panel by a selector valve handle.

3.9.5 Operator's control panel. Control levers, throttles, valves, and gages needed for, or pertaining to, the rewind operation shall be incorporated into, or located near, a weatherproof control panel. The control panel shall be so located that the operator will have an unobstructed view of the runway.

3.9.6 Purchase element tape cleaning device. A cleaning device shall be provided for the removal of stones, cinders, and all other runway debris from each purchase element tape before the tapes are rewound on the tape reels. The cleaning device shall neither cause excessive wear or heat damage to the tapes, nor shall it compromise arresting system performance. Disposal of the accumulated debris removed from the tapes shall be facilitated by suitable ports which will permit convenient access to the debris.

3.10 Purchase element tapes. Purchase element tapes (purchase tapes) shall be in accordance with MIL-W-38461, type III. The purchase element tape length will be variable due to system runout, runway span, and the distance between the runway edge sheaves and energy absorber. Purchase element tape length shall be as specified for each installation (see 6.2). The tapes shall be of sufficient length to allow 950 feet of aircraft runout after allowance has been made for the separation distance between the runway edge sheaves and the tape reels. The tapes shall have sewed end-loops for attachment to the tape connectors.

3.11 Purchase element tape connectors. Purchase element connectors for connecting the purchase element tapes to the hook cable shown on Drawing 52-E-182 shall be replaced with a connector conforming to Drawing 510445.

3.12 Hook cable. The hook cable (arresting pendant) shall be 1-1/4 inch diameter (12 by 6) (6 by 30) high strength, steel, preformed, nonrotating, wire rope with independent polypropylene core in accordance with NAEC Misc. No. 08787. The hook cable assembly shall be in accordance with Drawing 515053 which shall be used in place of the pendant assembly shown on Drawing 52-E-182. The hook cable length will be variable depending on the runway span and shall be as specified (see 6.2).

3.13 Runway edge sheaves. The runway edge sheave (deck sheave) assemblies shall be designed for rapid installation on a graded soil surface, but also shall be compatible with installation on concrete foundations. The sheave assemblies shall withstand loads up to tape breaking strength at 180° angle of tape wrap. The sheave assemblies shall accommodate bidirectional engagement of the arresting system.

3.14 Fairlead tubes and couplings. Fairlead tubes and couplings used between runway edge sheaves and the energy absorber to protect purchase element tapes from the effect of weather normally will be furnished by

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the Government. When specified (see 6.2), the Fairlead tubes and couplings shall be furnished by the contractor. Fairlead tubes and couplings normally shall be 12-inch outside diameter transite tubing, or equal. The length of tubing required is roughly equivalent to the sum of the split distances between the energy absorbers and runway edge sheaves.

3.15 Special service tools and kits. Unless otherwise specified (see 6.2), special service tools and kits for installation and maintenance of the barrier, arresting system shall be supplied with the system. The special service tools and kits shall contain at least the following:

a. Spare parts kits

b. Special wrenches

c. Stake driving tools (for soil surface installations only)

d. Other items as specified by the procuring activity (see 6.2).

3.16 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.17 Electromagnetic interference characteristics. Electromagnetic interference control requirements shall be in accordance with MIL-STD-826, method 7001.

3.18 Screw threads. Screw threads shall conform to FED-STD-H28.

3.19 Finishes and protective coatings. Cleaning, painting, plating, chemical films, and anodic coatings shall be in accordance with MIL-STD-808. All painted surfaces shall be painted for type I exposure. The colors shall be as follows:

a. Exterior surfaces - Color code DG (yellow)

b. Interior surfaces - Color code LG (aircraft gray)

c. Equipment within enclosure - Color code LG (aircraft gray).

3.20 Markings

3.20.1 Nameplates, instruction plates, data plates, and placards. All nameplates, instruction plates, data plates, and placards shall conform to MIL-P-514 and shall be of a size and shape consistent with the information required. Abbreviations shall be in accordance with MIL-STD-12. Mechanical symbols shall be in accordance with MIL-STD-17. All such plates shall be attached by metal screws or rivets.

3.20.1.1 Vendor's nameplate. Only one vendor's nameplate, not exceeding 6 by 9 inches in size, shall be installed on the arresting system.

3.20.2 Operational markings. The operational markings shall be applied in accordance with the marking requirements of MIL-STD-808. Stenciling shall be in accordance with MIL-STD-808.

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3.20.3 Identification of controls and instruments. Where applicable, all gages, controls, and instruments used in the operation of the arresting system shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light, et cetera, will not cause them to fade or become eradicated. Tags shall not be used.

3.20.4 Operating instructions. Where applicable, brief operating and precautionary instructions shall be permanently affixed on or near the control panel or other points as necessary. The instructions shall be clear, concise, and adequate to enable operation without damage to the system or injury to operating personnel. Precautionary instructions shall be so designed and located that the operator cannot fail to notice them before controls, valves, and switches are operated.

3.21 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.22 Workmanship. The aircraft arresting system, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of all parts, and tightness of assembly screws and bolts.

3.22.1 Cleaning. The aircraft arresting system shall be thoroughly cleaned and loose, spattered, or excess solder; metal chips; and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the aircraft arresting system shall be classified as follows:

a. First Article testing . . . . . See 4.3

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b. Acceptance tests . . . . . .See 4.4.

4.3 First Article testing (see 6.2)

4.3.1 Test sample. One barrier aircraft arresting system shall be subjected to the tests specified in 4.3.3.

4.3.2 Test report. After completion of first article testing, a test report shall be prepared in accordance with MIL-STD-831 and three copies furnished to the procuring activity.

4.3.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, and miles or position in the test procedure, as appropriate. Test conditions at the time of the events identified shall be recorded

b. A summary of the engineering analysis and tests conducted to determine assignable causes for any failure or irregular functioning

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control or other procedures, or leading to any corrections or changes proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such operations

d. Clock time and man-hours required for each maintenance and servicing action during the tests. A brief description of the experience and qualifications of the personnel taking such actions shall be included and shall be adequate to permit comparison to the personnel anticipated in similar field work

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.3.3 First Article tests. First article tests shall consist of all the tests described under 4.5.

4.3.3.1 First Article testing by the contractor. Unless otherwise specified (see 6.2), one barrier aircraft arresting system shall be subjected to all tests under 4.5, except 4.5.4, 4.5.6, and 4.5.7. The arresting system then shall be submitted to the procuring activity for the tests specified in 4.5.4, 4.5.6, and 4.5.7.

4.4 Acceptance tests. The acceptance tests shall consist of the individual tests.

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4.4.1 Individual tests. Each aircraft arresting system shall be subjected to the following tests as described under 4.5:

a. Examination of product . . . . . .See 4.5.1

b. Mechanical inspection . . . . . .See 4.5.2

4.5 Test methods

4.5.1 Examination of product. The aircraft arresting system shall be inspected to determine compliance with the requirements of this specification. All features and components shall be inspected with respect to materials, workmanship, dimensions, proper installation, and marking.

4.5.2 Mechanical inspection. A critical mechanical inspection shall be made of all components and parts of the aircraft arresting system to insure proper functioning of all mechanical parts, accessibility of controls and adjustments, installation, and to demonstrate interchangeability of removable parts without special matching. All components shall be moved through their maximum normal displacement and shall exhibit no malfunctioning, leakage, objectionable rough operation, or irregular movements. The inspection shall include complete lubrication and servicing of the energy absorber unit to insure readiness for operation.

4.5.3 Environmental tests. The aircraft arresting system shall be subjected to the following tests conducted in accordance with the specified methods of MIL-STD-810:

a. High temperature - method 501, procedure I, except the temperature shall be 125°F

b. Low temperature - method 502, procedure I, except the temperature shall be -40°F

c. Humidity - method 507, procedure I, except the temperature shall be 125°F

d. Fungus - method 508, procedure I

e. Salt-fog - method 509, procedure I

f. Rain - method 506, procedure I

g. Dust - method 510, procedure I, except the temperature shall be 125°F.

4.5.4 Operational test (to be conducted by the procuring activity). The aircraft arresting system shall be functionally tested against manned aircraft to determine compliance with the requirements specified herein. These tests shall be conducted over a full range of aircraft weights and speeds as specified in 3.8.2 and shall include engagements made on-center and off-center up to 50 feet.

4.5.4.1 Energy absorber temperature. The energy absorber shall be engaged at short-cycle intervals for 20 high-energy arrestments in 1 hour during which time no critical energy absorber component shall exceed a temperature which would deteriorate or degrade system performance. The energy absorber shall be

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capable of meeting the requirements specified in 3.8.

4.5.5 Electromagnetic interference. The aircraft arresting system shall be tested for electromagnetic interference, through all operations of a complete arrestment cycle, in accordance with MIL-STD-826, method 7001.

4.5.6 Reliability demonstration and testing (to be conducted by the procuring activity). The entire arresting system shall be installed and tested under operational conditions at a designated test site or operational base. This testing will be accomplished by the procuring activity. However, on-site technical support personnel shall be provided when specified (see 6.2).

4.5.7 Maintainability evaluation (to be conducted by the procuring actviity). The aircraft arresting system shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, etc., that may hinder servicing and operation. The testing will be accomplished by the procuring activity. However, on-site technical support personnel shall be provided when specified (see 6.2).

* 4.6 Inspection of packaging. Preservation, packaging, packing and marking shall be inspected to determine conformance to the requirements of Section 5, herein.

* PACKAGING

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

* 5.1.1 Level A

* 5.1.1.1 Disassembly. Disassembly of the system components shall be accomplished to the extent practicable to reduce cube and facilitate packaging and packing of the system.

* 5.1.1.2 Cleaning and drying. Component parts of the aircraft arresting system shall be cleaned in accordance with MIL-P-116, process C-1, and dried in accordance with MIL-P-116, procedure D-1 or D-4.

* 5.1.1.3 Preservation application. Immediately after cleaning and drying, all exposed bare metal surfaces and critical parts (i.e., sprockets, gears, chain drives, etc) shall be coated with P-1 or P-2 preservative in accordance with MIL-P-116, method I. Where ease of removal is not a prime consideration, P-1 preservative should be used.

* 5.1.1.4 Engines. Engines shall be preserved and packaged in accordance with MIL-E-10062.

* 5.1.1.5 Electrical components. Electrical components shall be preserved in accordance with MIL-E-17555.

* 5.1.1.6 Instrument panel. Glass on the instrument panel shall be covered with fiberboard conforming to PPP-F-320 or other similar protective material and secured with tape conforming to A-A-1830.

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* 5.1.1.7 Hook cable. The hook cable (pendant) shall be preserved and packaged in accordance with MIL-STD-163.

* 5.1.1.8 Purchase element tapes. Purchase element tapes shall be packaged in accordance with MIL-W-38461, except as modified in 5.2.1.2 herein.

* 5.1.1.9 Tools. Special service tools shall be preserved and packaged in accordance with PPP-P-40.

* 5.1.1.10 Kits. All kits supplied with the system shall be preserved and packaged in accordance with MIL-STD-2073-1, appendix D.

* 5.1.2 Level C. The level C preservation for the aircraft arresting system shall conform to the MIL-STD-2073-1 requirements for this level.

* 5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Components of the system, preserved and packaged as specified in 5.1, shall be packed as specified in 5.2.1.1 through 5.2.1.9.

* 5.2.1.1 Energy absorber. Each energy absorber, including the purchase element tape cleaning device, shall be packed in a container conforming to MIL-C-104, type I, class 2, style a.

* 5.2.1.2 Purchase element tapes. Purchase element tapes shall be packed in accordance with MIL-W-38461, except reels shall be modified to provide for skids and rubbing strips to provide ease of forklift entry.

* 5.2.1.3 Hook cables. Hook cables (pendants) shall be packed in accordance with MIL-STD-163.

* 5.2.1.4 Purchase element connectors. Purchase element connectors shall be packed in containers conforming to PPP-B-601, overseas type, style A or B.

* 5.2.1.5 Runway edge sheaves. Runway edge sheaves shall be packed in containers conforming to PPP-B-601, overseas type, style A or B.

* 5.2.1.6 Fairlead tubes and couplings. Fairlead tubes and couplings when specified (see 6.2), shall be packed in containers conforming to PPP-B-621, class 2.

* 5.2.1.7 Special service tools. Tools shall be packed in accordance with PPP-P-40.

* 5.2.1.8 Kits. Kits supplied with the system shall be packed in accordance with MIL-STD-2073-1, appendix D.

* 5.2.1.9 Consolidation, limitations, and closure. Components shall be consolidated into the fewest number of crates practicable consistent with required protection. The gross weight of each pack shall not exceed weight and size limitations specified in the applicable container specification. Containers shall be closed and strapped in accordance with the applicable container specification and appendix thereto.

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* 5.2.2 Level B. Level B packing shall be the same as level A, except domestic type containers, where available, in accordance with the applicable container specification shall be used in lieu of overseas type containers.

* 5.2.3 Level C. The level C packing for the aircraft arresting system shall conform to the MIL-STD-2073-1 requirements for this level.

* 5.2.4 Palletized loads. When specified (see 6.2) palletized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.3 Marking.

* 5.3.1 Levels A, B, and C. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. In addition, to facilitate reassembly, mating parts that are removed from the basic assembly shall be matchmarked with lusterless white enamel and overcoated with water-resistant clear spar varnish conforming to TT-V-121. Small parts that cannot be matchmarked as specified above, shall be matchmarked with waterproof tags conforming to UU-T-81, Type A, or B, Class 2.

* 5.4 General.

* 5.4.1 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.1.

* 5.4.2 Physical protection. Interior blocking, bracing, cushioning, et cetera, shall be in accordance with MIL-STD-1186.

6. NOTES

6.1 Intended use. The BAK-13A/F48A barrier aircraft arresting system is intended for operational and emergency arrestment of hook equipped aircraft at speeds up to 180 knots at energy levels up to 85 million foot-pound for a 950-foot system runout.

* 6.2 Ordering data.

* 6.2.1 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. In the event it becomes necessary to change the drawings (see 3.3) in order to achieve the performance requirements of the specification, the contractor will immediately notify the contracting officer with recommendations for drawing changes to obtain the required performance. To the extent approved, such changes will be processed in accordance with the provisions of the changes clause of the contract

c. Type of installation for which each arresting system is to be used (see 3.6, 3.6.1, 3.6.2, 3.6.3) including

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(1) Length of hook cable (see 3.12)

(2) Span between runway edge sheaves

(3) Split distance between runway edge sheaves and the energy absorber

(4) Item, components, or foundations that will be furnished by the government

d. For soil surface installations (see 3.6.1)

(1) When reinstallation kit is to be supplied (see 3.6.1.1)

e. For permanent, surface type concrete foundations installations (see 3.6.2)

(1) When concrete foundations are to be provided

(2) When Fairlead tubes and couplings are to be supplied by the contractor (see 3.14)

f. For permanent, pit type installations (see 3.6.3)

(1) Drawings and specifications required, and listing of any government-furnished items, components, or foundations

(2) When Fairlead tubes and couplings are to be supplied by the contractor (see 3.14)

g. Purchase element tape length for each arresting system (see 3.10)

h. When special service tools and kits are not to be supplied with the arresting system (see 3.15)

i. Any special items to be included in the special service tools and kits (see 3.15)

j. Location and conditions for preproduction testing (see 4.3) including:

(1) Tests to be conducted by the contractor (see 4.3.3.1)

(2) Tests to be conducted by the procuring activity (See 4.3.3.1)

(3) Extent of contractor support and participation during preproduction testing by the procuring activity (see 4.5.4 and 4.5.6, and 4.5.7)

k. Applicable levels of preservation, packing and packaging (see 5.1 and 5.2).

* l. When palletized loads are required (see 5.2.4).

6.3 Definitions

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified number of cycles.

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6.3.2 Runway span. Runway span is the cross-runway distance between the systems runway edge sheaves. This distance does not necessarily coincide with the runway width.

6.3.3 Split distance. Split distance is the distance between the runway edge sheave and the energy absorber on a given side. This distance is measured from the geometric center of the runway edge sheave to the edge of the energy absorber nearest the runway edge sheave.

* 6.4 First article. When a first article inspection is required, the item(s) should be a standard production item. The first article should consist of (one) unit(s). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

* 6.5 Subject term (keyword) listing.

Aircraft Arresting, Barrier
Arresting system, Aircraft
Barrier, Aircraft Arresting
BAK-14A/F48A Aircraft arresting system
Hydraulic, Barrier Arresting

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-83183B(USAF)	BARRIER AIRCRAFT ARRESTING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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